YEAR 7 · UNIT 4 REVIEW · SCIENCE UNIT 3 · HOMEWORK 10

Faster in January

Part 1: Maths — Unit 4 review
Part 2: Science — 3.2, 3.3 and the football model, with decimals

Name: _____ **Class:** _____

Date given: _____ **Date due:** Friday 30 October 2026

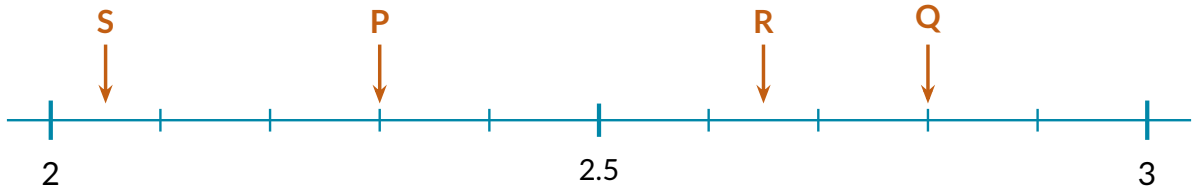
How to do this homework

- **Sections A–C:** maths (Unit 4). **Sections D–F:** science. Section F uses decimals.
- **No calculator. Estimate first.** Then write the **calculation**, then the answer.
- Answer in **full sentences** when it says so. *"Gravity"* is not a sentence. *"Gravity pulls the planet towards the Sun"* is.

Section A · Unit 4 Review — Comparing, Adding and Subtracting

A1 Read the line

The line goes from **2** to **3**. Every small line is **0.1**. **(a)** Write the number each arrow points to.



P _____	Q _____	R _____	S _____

(b) Write the four numbers in order. Start with the **largest**.

_____

A2 Which is bigger?

Write <, > or = in each box.

Tip: compare the whole numbers first. Then **add zeros**: 0.7 = 0.70, so 0.7 > 0.65.

(a) 0.5 ☐ 0.45

(b) 2.1 ☐ 2.09

(c) 8.08 ☐ 8.8

(d) 6.4 ☐ 6.400

(e) 15.6 ☐ 9.87

(f) 1.25 ☐ 1.3

A3 Line up the points

Written method

Write the numbers with the **points in a line**. Fill the gaps with **zeros**. The point in the answer goes **straight down**.

(a) 7.8 + 4.56

(b) 23.4 + 0.87 + 5

(c) 16.25 + 3.9

(d) 10 − 3.42

(e) 6.3 − 2.75

(f) 0.9 − 0.36

Section B · Unit 4 Review — Multiplying and Dividing

Where the point goes

Multiply Take the point out and multiply. Then **estimate** to put it back.
Example: 3.8 × 12 → 38 × 12 = 456 → estimate 4 × 12 = 48 → **45.6**

Divide Short division. The point goes **straight up**. If a digit is too small to divide, write **0**. Dividing by a 2-digit number? Write its **times table** first.

B1 Point out, point back

Fill in the table. Row 1 is done for you.

Calculation	Estimate	Multiply without the point	Answer
2.4 × 13	2 × 13 = 26	24 × 13 = 312	31.2
0.6 × 9			
3.45 × 6			
7.2 × 15			
2.8 × 23			

B2 The point goes straight up

Use short division in each box. For **(e)** and **(f)**, write the 15 and the 24 times tables first.

(a) 9.45 ÷ 5

(b) 4.32 ÷ 4

(c) 0.72 ÷ 9

(d) 7.3 ÷ 2

(e) 52.5 ÷ 15

(f) 81.6 ÷ 24

Section C · Unit 4 Review — Words, Quick Ways and Money

C1 From English to a calculation

Word help — the words that choose the operation

each	for one of them. *3 pens at 2 dollars each*: 3 × 2
altogether	all of it, the total
twice as far	2 times as far
What must you add to 4 to make 7?	7 − 4. It says *add*, but you **subtract**
3 less than 10	10 − 3. The **second** number comes **first**

Write the **calculation first**. Then write the answer, with its unit if it has one (m, km).

(a) What must you add to 4.65 to make 7?

Calculation: _____ **Answer:** _____

(b) A ribbon 2.4 m long is cut into 16 equal pieces. How long is each piece?

Calculation: _____ **Answer:** _____

(c) Mr Bowen walks 2.5 km on Monday. On Tuesday he walks twice as far. How far does he walk altogether?

Calculation: _____ **Answer:** _____

(d) What is 3.5 less than 10?

Calculation: _____ **Answer:** _____

C2 Now you write the question

Write a word problem for $12 \times 2.40 = 28.80$

Your word problem must have: tick each one

- ☐ a real situation
- ☐ a word from C1
- ☐ full sentences
- ☐ units (dollars, kg)
- ☐ a question at the end
- ☐ the answer 28.80

C3 Use a quick way

Use the **same quick way** as the example. Write the quick way, then the answer.

Quick way	Example	Your turn
× 9: × 10, then take away one lot	$3.6 \times 9 = 36 - 3.6 = 32.4$	(a) $4.7 \times 9 =$ _____ = _____
× 11: × 10, then add one lot	$2.3 \times 11 = 23 + 2.3 = 25.3$	(b) $6.2 \times 11 =$ _____ = _____
Split the number	$4.3 \times 6 = 24 + 1.8 = 25.8$	(c) $5.3 \times 4 =$ _____ = _____
× 0.25 = a quarter of	$28 \times 0.25 = 28 \div 4 = 7$	(d) $64 \times 0.25 =$ _____ = _____
× 0.07 = ÷ 100, then × 7	$200 \times 0.07 = 2 \times 7 = 14$	(e) $300 \times 0.07 =$ _____ = _____
× 0.5 = half of	$7.4 \times 0.5 = 7.4 \div 2 = 3.7$	(f) $9.6 \times 0.5 =$ _____ = _____

C4 Change from 10 dollars

Money has **2 decimal places**: 7.8 dollars is written 7.80 dollars.

Mr Bowen buys 3 bánh mì at 1.35 dollars each and 2 iced coffees at 1.90 dollars each. He pays with a 10-dollar note. How much change does he get?

1. 3 bánh mì: _____ dollars

2. 2 iced coffees: _____ dollars

3. Total: _____ dollars

4. Change: _____ dollars

Working

C5 Mr Bowen's homework

Only **one** of Mr Bowen's answers is right. Write **right** or **wrong** for each one. If it is wrong, write the correct answer.

Mr Bowen wrote	Right or wrong?	Correct answer
(a) 4.5 + 2.7 = 6.12		
(b) 9.6 ÷ 3 = 3.2		
(c) 3.40 > 3.4		
(d) 60 × 0.4 = 2.4		

Section D · Science 3.2 — How the Solar System Formed

D1 Key words

Match each word to its meaning. Write the letter (A–F) in the box.

nebula ☐
hypothesis ☐
evidence ☐
model ☐
limitation ☐
plane ☐

A an idea that can be tested
B one flat surface
C a cloud of dust and gas in space
D facts from observations or experiments
E something a model cannot do, gets wrong or leaves out
F a way of showing something that is hard to observe directly

D2 Support or contradict?

The **hypothesis**: **The Sun and the planets formed from one spinning cloud of dust and gas.**

Does each fact **support** it (agree) or **contradict** it (go against it)? Tick **one** box.

Fact	supports	contradicts
All the planets orbit the Sun in the same direction.		
All the orbits are in the same plane.		
Venus spins the opposite way to most planets.		
HL Tauri is a young star with a flat disc of dust round it.		
Uranus spins on its side.		
Most moons orbit their planet in the same direction.		

(b) Two facts contradict the hypothesis. Is the hypothesis wrong? Answer in a full sentence. Use the words **most of the evidence**.

D3 From a cloud to the Sun and planets

(a) Put the four steps in order. Write **1**, **2**, **3** and **4** in the boxes.

☐ The hot middle becomes the Sun.

☐ Gravity pulls the nebula together. It spins faster.

☐ Dust sticks together and grows into planets.

☐ The nebula flattens into a spinning disc.

(b) Why do all the planets orbit the Sun in the same direction? Answer in a full sentence. Start: **"They all formed from …"**

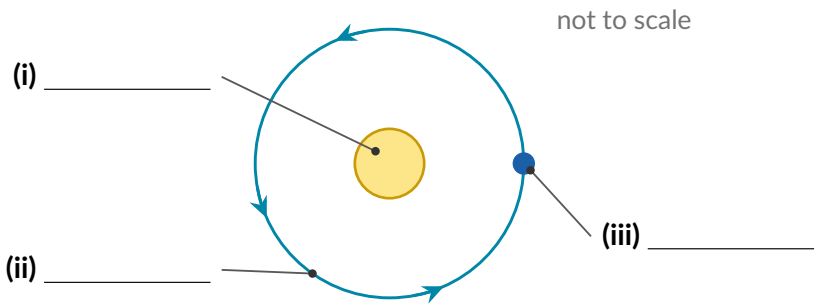
(c) Mercury, Venus, the Earth and Mars are small and rocky. Why? Start: **"Near the Sun, it was too hot for …"**

(d) The Sun has **99.8%** of the Solar System's mass. What % is left for the rest? _____ %

Section E · Science 3.3 — Movement in Space

E1 What keeps a planet in orbit?

(a) Label the drawing. Use these words: **Sun** · **orbit** · **planet**



(b) Draw an **arrow** on the planet. Show which way the Sun's **gravity** pulls it.

(c) The arrows show which way the planet moves. Now the Sun's gravity **switches off**. Draw a **dashed line** to show where the planet goes.

E2 Which forces?

Which forces act on each one? Tick **every** box that is true. Some rows need **two** ticks.

	gravity	air resistance
The Earth, orbiting the Sun		
The Moon, orbiting the Earth		
A ball falling from the balcony		
A plane landing on a runway		

(b) Why do the planets never slow down? Start: **"Space is a …, so …"**

Section F · Science in Decimals — Faster Near the Sun

Speed, from Science 3.3

Speed: how fast something moves. **km/s** = kilometres **per** second.
As the distance from the Sun **increases**, the speed of the planets **decreases**. The Sun's pull is **weaker** far away.

F1 How much faster? How many times as fast?

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
speed (km/s)	**48**	**35**	**30**	**24**	**13**	**10**	**7**	**5**

Fill in the table. Row 1 is done for you. In each row, the **first** planet is the faster one.

Planets	**How much faster?** **subtract**: the difference	**How many times as fast?** **divide**: big ÷ small
Mercury and Neptune	48 − 5 = 43 km/s	48 ÷ 5 = 9.6
Mercury and the Earth		
Mars and Saturn		
The Earth and Mars		

F2 The Sun is a football

In the lesson the Sun became a football, and everything else shrank by the same amount.

	the Moon	the Earth	Neptune	Jupiter	the Sun
Model width	0.5 mm	2 mm	7.8 mm	2.3 cm	22 cm
Like	a grain of sand	a pinhead	a pea	a grape	a football

(a) Write the five widths in order, **smallest** first. Remember: **1 cm = 10 mm**.

_____

(b) Mr Bowen puts the football against one wall of his classroom, which is **8.5 m** long. The Earth must be **24 m** from the football. How far past the other wall is the Earth?



Calculation: _____ **Answer:** _____

(c) Give **one limitation** of the model. Start: **"One limitation is …"**

_____

_____

F3 Faster in January

The Earth's orbit is *almost* circular. In January the Earth is a little **closer** to the Sun than in July.

	Distance from the Sun	Speed
January	147.1 million km	30.3 km/s
July	152.1 million km	29.3 km/s

(a) How much closer to the Sun is the Earth in January?

Calculation: _____ **Answer:** _____

(b) In which month is the Earth faster? Why? Use the words **closer** and **pull**. Start: **"The Earth is faster in …, because …"**

_____

_____

(c) How much faster is the Earth in January than in July?

Calculation: _____ **Answer:** _____

(d) Find the **mean** of the two speeds.

Calculation: _____ **Answer:** _____

(e) Round your mean to the nearest whole number. _____

(f) In the lesson, the Earth's speed was **30 km/s**. What do you notice?

_____

_____